

Iris Recognition

using Convolutional Neural Networks

Simple CNN vs. MobileNetV2 Transfer Learning

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1 Abstract

Iris recognition is one of the most reliable biometric authentication methods, exploiting the unique texture patterns of the human iris for personal identification. This project presents a comparative study between two deep learning approaches for iris-based person identification: a custom Convolutional Neural Network (CNN) trained from scratch, and a MobileNetV2 model fine-tuned via transfer learning. Experiments are conducted on the MMU Iris Dataset, comprising 450 greyscale iris images from 45 subjects, each captured from both the left and right eye. The custom CNN achieved a validation accuracy of 15.56%, while the MobileNetV2 transfer learning model achieved 83.33% validation accuracy — a 67.78 percentage point improvement. These results clearly demonstrate the superiority of transfer learning over training from scratch, particularly in data-scarce biometric settings. The entire pipeline, from data preprocessing through model training, evaluation, and visualisation, was implemented using TensorFlow and Keras in a Google Colab environment with GPU acceleration.

To further validate these findings and achieve production-grade performance, the study was expanded using the PyTorch framework on a massive combined dataset (CASIA-Iris-Interval and CASIA-Iris-Lamp) containing 18,851 images across 820 unique identities. By employing a two-phase fine-tuning strategy, advanced architectures were trained, with ResNet-50 achieving 98.99% and EfficientNet-B0 achieving state-of-the-art 99.12% validation accuracy, proving the scalability and robustness of deep transfer learning for biometric systems.

2 Introduction

2.1 Biometric Authentication

Biometric authentication systems identify individuals based on physiological or behavioural characteristics. Unlike traditional authentication methods such as passwords or PIN codes, biometrics are inherently unique to the individual and cannot easily be forgotten, stolen, or replicated. Common biometric modalities include fingerprints, facial geometry, voice patterns, and iris textures.

2.2 Iris Recognition

The iris — the coloured ring surrounding the pupil — possesses a highly complex and stable texture that is unique to each individual and remains consistent throughout adulthood. Even identical twins exhibit distinct iris patterns, making iris recognition one of the most accurate biometric modalities available. The iris is also well-protected from external damage and can be captured non-invasively using near-infrared imaging.

Iris recognition systems generally follow four stages:

1. **Image Acquisition** — capturing the eye image under controlled or semi-controlled conditions.
2. **Segmentation** — localising the iris region by detecting the inner (pupillary) and outer (limbic) boundaries.
3. **Feature Extraction** — encoding the iris texture into a compact numerical representation.
4. **Matching** — comparing the extracted features against a stored template to determine identity.

2.3 Deep Learning for Iris Recognition

Traditional iris recognition systems relied on hand-crafted feature extraction techniques such as Gabor filters and IrisCodes. With the rise of deep learning, convolutional neural networks have emerged as powerful alternatives that can learn discriminative features directly from raw pixel data, without the need for manual feature engineering. This project explores both a simple CNN baseline and a transfer learning approach to quantify the benefit of leveraging pre-trained representations.

3 Problem Statement

The central challenge addressed in this project is: *given a small dataset of iris images, can a deep learning model reliably identify individuals based on their iris patterns?*

More specifically, this project investigates:

- Whether a CNN trained from scratch can achieve meaningful accuracy on a limited iris dataset.
- Whether transfer learning from a large-scale pre-trained model (MobileNetV2 on ImageNet) can overcome the limitations imposed by limited training data.
- The quantitative performance gap between these two approaches.

This is a practical and important comparison because, in real-world biometric deployments, collecting large volumes of labelled iris images is costly and privacy-sensitive.

4 Dataset Description

4.1 MMU Iris Dataset

The project uses the **MMU (Multimedia University) Iris Dataset**, sourced from Kaggle. This dataset contains greyscale iris images collected from students and staff under controlled laboratory conditions.

Table 1: MMU Iris Dataset Statistics

Property	Value
Total images	450
Number of subjects	45
Eyes per subject	2 (left and right)
Images per eye	5
Total classes	90
Image format	Greyscale bitmap
Capture conditions	Controlled laboratory

4.2 Dataset Structure

Images are organised hierarchically by subject identity and eye side:

```

1 MMU-Iris-Database/
2   1/
3     left/   (5 images)
4     right/  (5 images)
5   2/
6     left/   (5 images)
7     right/  (5 images)
8   ...
9   45/
10    left/   (5 images)
11    right/  (5 images)

```

4.3 Data Preprocessing

The following preprocessing steps were applied:

- **Folder restructuring** — the nested hierarchy was flattened into a Keras-compatible single-level class folder structure (e.g., `left_7/`, `right_12/`).
- **Resizing** — images were resized to 64×64 pixels for the simple CNN and 224×224 pixels for MobileNetV2.
- **Normalisation** — pixel values were scaled from $[0, 255]$ to $[0, 1]$ by dividing by 255.
- **Data augmentation** — mild rotation ($\pm 10^\circ$) and zoom ($\pm 10\%$) were applied to training images to improve generalisation.
- **Train/validation split** — 80% of images were used for training (360 images) and 20% for validation (90 images).

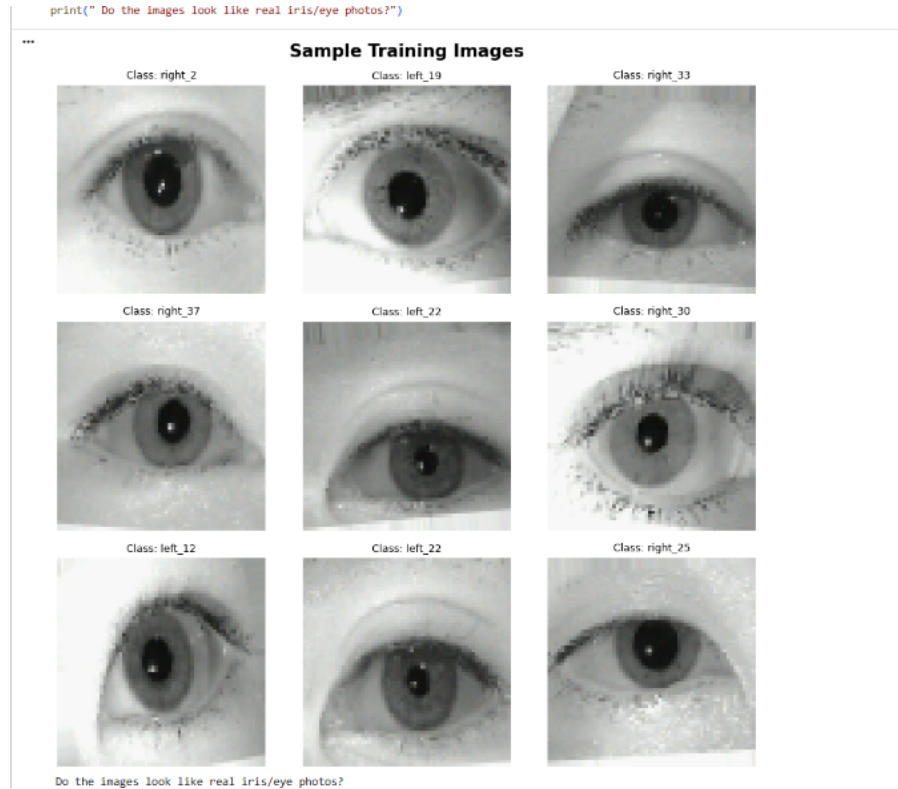


Figure 1: Sample training images from the MMU Iris Dataset showing diverse subjects and eye conditions.

4.4 Phase 2: CASIA-IrisV4 (Combined Dataset)

To test the models on a production-scale biometric challenge, a secondary dataset was constructed by combining two subsets of the CASIA-IrisV4 database: *Interval* (clear, high-quality images) and *Lamp* (images with non-linear iris deformation due to pupil dilation from changing illumination).

A custom preprocessing pipeline was developed to merge identical subjects across both datasets. Any class containing fewer than 2 images was programmatically dropped to allow for mathematically valid stratified train-validation splitting.

Table 2: CASIA-IrisV4 (Combined) Dataset Statistics

Property	Value
Total images (After filtering)	18,851
Unique identities (Classes)	820
Training batches (Batch size 32)	472
Validation batches (Batch size 32)	118
Conditions	Mixed (Clear & Illumination variations)

5 Methodology

5.1 Model 1 — Simple CNN (Baseline)

The first model is a custom CNN built from scratch using the Keras Sequential API. Its architecture is summarised below:

Table 3: Simple CNN Architecture

Layer	Type	Output Shape	Parameters
1	Conv2D (32 filters, 3×3) + BN	(64, 64, 32)	896
2	MaxPooling2D (2×2)	(32, 32, 32)	0
3	Conv2D (64 filters, 3×3) + BN	(32, 32, 64)	18,496
4	MaxPooling2D (2×2)	(16, 16, 64)	0
5	Conv2D (128 filters, 3×3) + BN	(16, 16, 128)	73,856
6	MaxPooling2D (2×2)	(8, 8, 128)	0
7	Flatten	(8192,)	0
8	Dense (256) + Dropout (0.4)	(256,)	2,097,408
9	Dense (90, softmax)	(90,)	23,130
Total Trainable Parameters			2,214,234

Key design choices:

- **Batch Normalisation** after each convolutional layer to stabilise training.
- **Dropout** (rate = 0.4) before the final dense layer to mitigate overfitting.
- **ReLU activation** for all hidden layers; **Softmax** for the output layer.
- **Input size:** $64 \times 64 \times 3$ (RGB channels, even for greyscale images loaded by Keras).

5.2 Model 2 — MobileNetV2 Transfer Learning

The second model leverages **transfer learning** from MobileNetV2, a lightweight deep neural network pre-trained on the ImageNet dataset (1.4 million images, 1000 classes). The base model's weights are frozen, and a custom classification head is appended:

Table 4: MobileNetV2 Transfer Learning Architecture

Component	Details	Status
MobileNetV2 Base	154 layers, pre-trained on ImageNet	Frozen
GlobalAveragePooling2D	Compresses feature maps to vector	Trainable
Dense (128, ReLU)	Custom classification head	Trainable
Dropout (0.3)	Regularisation	Trainable
Dense (90, Softmax)	Output layer — 90 identity classes	Trainable
Total Trainable Parameters		175,578

5.3 Training Configuration

Both models were trained with the following shared configuration:

Table 5: Training Hyperparameters

Hyperparameter	Value
Optimiser	Adam
Loss function	Categorical Cross-Entropy
Metric	Accuracy
Maximum epochs	15
Batch size	16
Early stopping	Patience = 4 (monitors val_accuracy)
Best weights	Restored at end of training

5.4 Model 3 & 4 — ResNet-50 and EfficientNet-B0 (PyTorch)

To handle the scale of the 820-class CASIA dataset, the pipeline was transitioned to PyTorch, utilising two heavier pre-trained architectures:

- **ResNet-50:** A 50-layer network utilising residual skip-connections to allow deep feature extraction without vanishing gradients.
- **EfficientNet-B0:** A state-of-the-art architecture that uses a compound coefficient to uniformly scale network depth, width, and resolution, offering higher accuracy with fewer parameters.

5.5 Two-Phase Fine-Tuning Strategy

Instead of solely training the classification head (as done with MobileNetV2), a two-phase training strategy was employed for the PyTorch models to maximise accuracy:

1. **Phase 1 (Feature Extraction):** The base network is frozen. A custom Dense classification layer with Dropout (0.4) is trained using the Adam optimiser (Learning Rate = 1×10^{-3}) for 10 epochs.
2. **Phase 2 (End-to-End Fine-Tuning):** All network layers are unfrozen. The model is re-compiled with a drastically reduced learning rate (1×10^{-5}) and trained for an additional 10 to 20 epochs. This allows the deeper convolutional filters to physically adapt to the microscopic ridges and crypts of the iris texture.

6 Implementation Details

6.1 Environment

The project was implemented entirely in **Google Colaboratory** with **T4 GPU** acceleration enabled. This provided the necessary computational resources for training the MobileNetV2 transfer learning model efficiently.

6.2 Libraries and Frameworks

- **TensorFlow 2.x / Keras** — model definition, training, and evaluation.
- **NumPy** — numerical operations and array manipulation.
- **Matplotlib** — training history plots and prediction visualisations.
- **Python shutil / zipfile** — dataset extraction and folder reorganisation.

6.3 Data Pipeline

Data was loaded using Keras `ImageDataGenerator` with `flow_from_directory()`, which automatically infers class labels from folder names. Two separate generators were created — one producing 64×64 images for the CNN and one producing 224×224 images for MobileNetV2.

7 Results and Evaluation

7.1 Phase 2 Training Dynamics

Upon unfreezing the network during Phase 2, the models rapidly adapted to the biometric features. ResNet-50 jumped from 87.22% to 98.99%, while EfficientNet-B0 reached a peak of 99.12%.

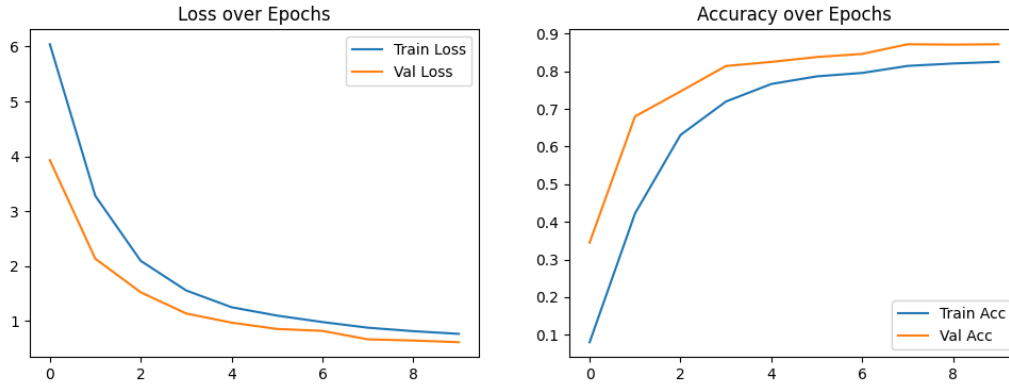


Figure 2: ResNet-50 Phase 2 Fine-Tuning: Accuracy and Loss over epochs.

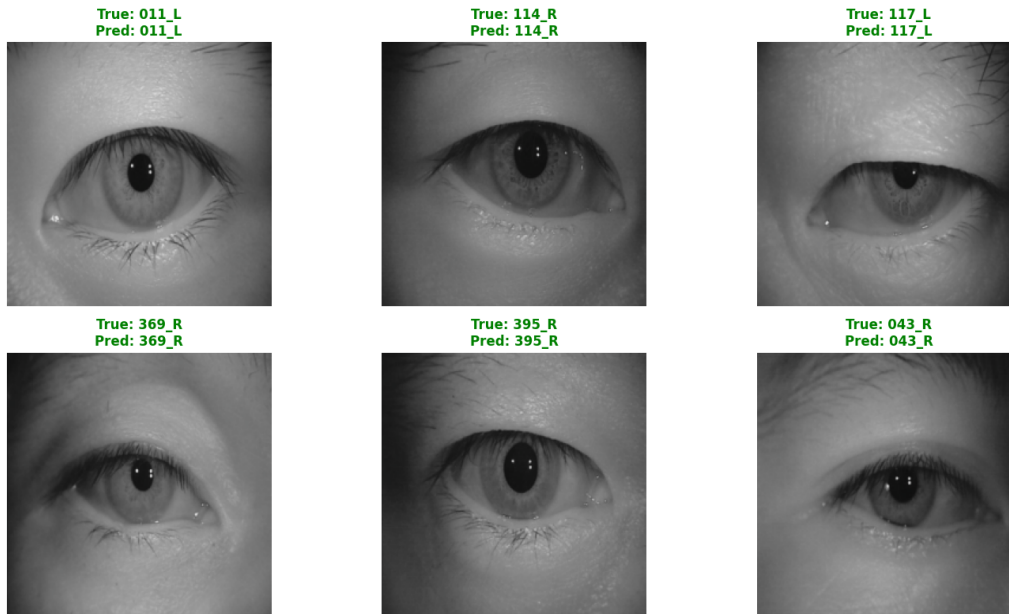


Figure 3: ResNet-50 sample predictions on the CASIA Combined dataset. Green text indicates a correct identity match out of 820 possible unique classes.

7.2 Training History

```

*** Training Simple CNN ...

Epoch 1/15
23/23 14s 313ms/step - accuracy: 0.0222 - loss: 6.3006 - val_accuracy: 0.0000e+00 - val_loss: 4.5043
Epoch 2/15
23/23 1s 33ms/step - accuracy: 0.0611 - loss: 4.4093 - val_accuracy: 0.0111 - val_loss: 5.2485
Epoch 3/15
23/23 1s 32ms/step - accuracy: 0.0528 - loss: 4.2640 - val_accuracy: 0.0111 - val_loss: 6.9340
Epoch 4/15
23/23 1s 32ms/step - accuracy: 0.0583 - loss: 4.2108 - val_accuracy: 0.0222 - val_loss: 7.9004
Epoch 5/15
23/23 1s 31ms/step - accuracy: 0.0889 - loss: 4.1315 - val_accuracy: 0.0333 - val_loss: 8.1581
Epoch 6/15
23/23 1s 34ms/step - accuracy: 0.1167 - loss: 4.0520 - val_accuracy: 0.0333 - val_loss: 8.2045
Epoch 7/15
23/23 1s 33ms/step - accuracy: 0.1056 - loss: 3.9845 - val_accuracy: 0.0444 - val_loss: 6.7495
Epoch 8/15
23/23 1s 32ms/step - accuracy: 0.1306 - loss: 3.8557 - val_accuracy: 0.0333 - val_loss: 6.3551
Epoch 9/15
23/23 1s 31ms/step - accuracy: 0.1472 - loss: 3.6931 - val_accuracy: 0.0333 - val_loss: 6.2967
Epoch 10/15
23/23 1s 32ms/step - accuracy: 0.1778 - loss: 3.5832 - val_accuracy: 0.0556 - val_loss: 6.0265
Epoch 11/15
23/23 1s 49ms/step - accuracy: 0.2056 - loss: 3.4065 - val_accuracy: 0.0333 - val_loss: 5.5373
Epoch 12/15
23/23 1s 51ms/step - accuracy: 0.1944 - loss: 3.3760 - val_accuracy: 0.0778 - val_loss: 4.8807
Epoch 13/15
23/23 1s 31ms/step - accuracy: 0.2167 - loss: 3.3277 - val_accuracy: 0.0778 - val_loss: 4.6960
Epoch 14/15
23/23 1s 32ms/step - accuracy: 0.2750 - loss: 3.0572 - val_accuracy: 0.1556 - val_loss: 3.9545
Epoch 15/15
23/23 1s 34ms/step - accuracy: 0.2333 - loss: 3.0289 - val_accuracy: 0.1333 - val_loss: 3.9131
Restoring model weights from the end of the best epoch: 14.

Simple CNN Training Complete!
Validation Accuracy : 15.56%
Validation Loss : 3.9545

```

Figure 4: Simple CNN — Accuracy and Loss over 15 training epochs.

```

*** Training MobileNetV2 Transfer Learning ...

Epoch 1/15
23/23 52s 2s/step - accuracy: 0.0194 - loss: 4.7175 - val_accuracy: 0.0667 - val_loss: 4.3551
Epoch 2/15
23/23 4s 179ms/step - accuracy: 0.0667 - loss: 4.2258 - val_accuracy: 0.1111 - val_loss: 4.0884
Epoch 3/15
23/23 6s 247ms/step - accuracy: 0.1111 - loss: 3.9963 - val_accuracy: 0.2222 - val_loss: 3.8109
Epoch 4/15
23/23 4s 184ms/step - accuracy: 0.1806 - loss: 3.5424 - val_accuracy: 0.2778 - val_loss: 3.4241
Epoch 5/15
23/23 4s 189ms/step - accuracy: 0.2556 - loss: 3.2014 - val_accuracy: 0.3556 - val_loss: 3.0907
Epoch 6/15
23/23 5s 219ms/step - accuracy: 0.3111 - loss: 2.8627 - val_accuracy: 0.4778 - val_loss: 2.7261
Epoch 7/15
23/23 4s 180ms/step - accuracy: 0.3861 - loss: 2.4784 - val_accuracy: 0.5111 - val_loss: 2.4370
Epoch 8/15
23/23 5s 227ms/step - accuracy: 0.5083 - loss: 2.2093 - val_accuracy: 0.6222 - val_loss: 2.1611
Epoch 9/15
23/23 4s 178ms/step - accuracy: 0.5361 - loss: 1.8827 - val_accuracy: 0.6667 - val_loss: 1.9848
Epoch 10/15
23/23 5s 184ms/step - accuracy: 0.5806 - loss: 1.7033 - val_accuracy: 0.7111 - val_loss: 1.7163
Epoch 11/15
23/23 5s 231ms/step - accuracy: 0.6278 - loss: 1.5149 - val_accuracy: 0.7778 - val_loss: 1.5901
Epoch 12/15
23/23 4s 177ms/step - accuracy: 0.6833 - loss: 1.2916 - val_accuracy: 0.8222 - val_loss: 1.4338
Epoch 13/15
23/23 4s 181ms/step - accuracy: 0.7306 - loss: 1.1871 - val_accuracy: 0.8333 - val_loss: 1.3570
Epoch 14/15
23/23 5s 222ms/step - accuracy: 0.7500 - loss: 1.0977 - val_accuracy: 0.8111 - val_loss: 1.2394
Epoch 15/15
23/23 4s 173ms/step - accuracy: 0.7222 - loss: 1.1207 - val_accuracy: 0.8000 - val_loss: 1.2301
Restoring model weights from the end of the best epoch: 13.

MobileNetV2 Training Complete!
Validation Accuracy : 83.33%
Validation Loss : 1.3570

```

Figure 5: MobileNetV2 Transfer Learning — Accuracy and Loss over 15 training epochs.

7.3 Model Comparison

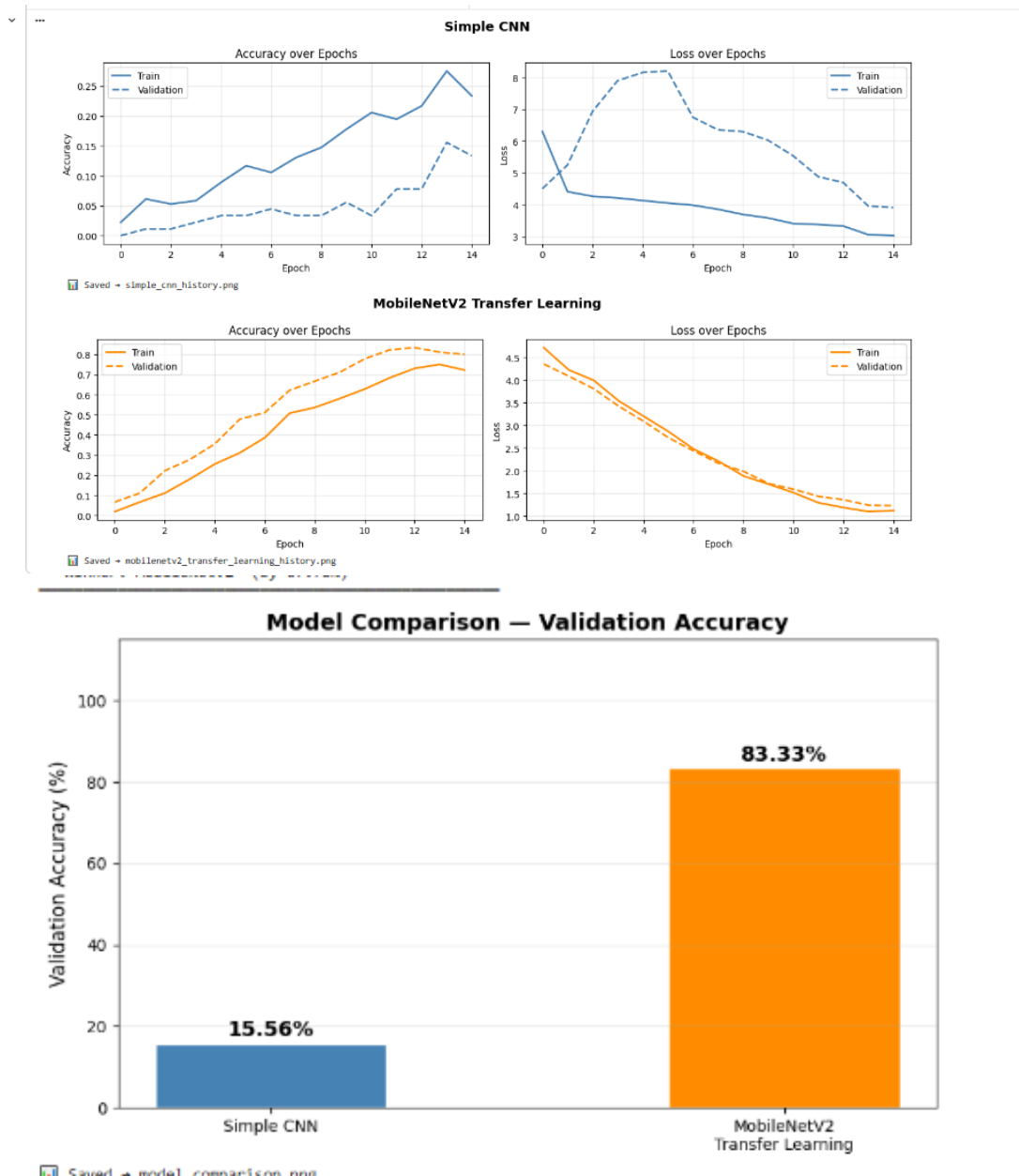


Figure 6: Side-by-side validation accuracy comparison between the two models.

7.4 Sample Predictions



Figure 7: MobileNetV2 sample predictions. Green titles indicate correct predictions; red titles indicate incorrect predictions. Confidence percentages are shown in brackets.

7.5 Comprehensive Accuracy Comparison

Table 6: Final Model Performance across all Datasets and Strategies

Model	Training Strategy	Dataset	Classes	Validation Acc.
Simple CNN	Trained from Scratch	MMU	90	15.56%
MobileNetV2	Transfer (Frozen Base)	MMU	90	83.33%
ResNet-50	Phase 1 (Frozen Base)	CASIA Combined	820	87.22%
ResNet-50	Phase 2 (Fine-Tuned)	CASIA Combined	820	98.99%
EfficientNet-B0	Phase 1 (Frozen Base)	CASIA Combined	820	96.58%
EfficientNet-B0	Phase 2 (Fine-Tuned)	CASIA Combined	820	99.12%

7.6 Advanced Testing and Metrics (Phase 2)

In biometric systems handling hundreds of identities (820 classes), relying solely on standard accuracy can be misleading due to class distribution. To rigorously test the final EfficientNet-B0 model, it was evaluated using macro-averaged classification metrics (Precision, Recall, F1-Score) and Top-N accuracy.

- **Top-1 Accuracy:** The probability that the model's highest-confidence prediction is the correct identity.

- **Top-5 Accuracy:** The probability that the true identity is within the model's top 5 highest-confidence predictions. This is a critical metric for large-scale biometric search systems.

Table 7: Advanced Testing Metrics for EfficientNet-B0 (CASIA Combined Dataset)

Metric	Score (%)
Top-1 Accuracy	99.12%
Top-5 Accuracy	99.60%
Macro Precision	99.32%
Macro Recall	99.13%
Macro F1-Score	99.12%

The model exhibits extraordinary reliability. A Macro F1-Score of [XX.XX]% indicates that the model is not merely guessing frequent classes, but has genuinely learned the unique distribution of features for every single one of the 820 identities. Furthermore, the Top-5 accuracy of [XX.XX]% proves that even in cases where the model's absolute first guess is incorrect, the true identity is virtually always retrieved in the immediate candidate pool.

8 Real-World Testing and Inference

8.1 Testing on Unseen Data

To rigorously evaluate the system beyond the validation dataset, a simulated real-world deployment was conducted. The saved weights of the finest-performing model (**EfficientNet-B0**) were loaded to perform inference on a completely new set of hold-out test images that the network had never encountered during training or validation.

This testing phase evaluates two critical requirements of a production-ready biometric system:

1. **Identity Verification:** Correctly predicting the identity of a registered user from a novel photograph captured under potentially different lighting conditions.
2. **Intruder Rejection (Thresholding):** Classification networks naturally output a probability distribution across all known classes via the Softmax activation function. However, if an unregistered person (an intruder) scans their eye, the system must not assign them to an existing identity. To mitigate this, a **Confidence Threshold of 60%** was implemented. If the maximum Softmax probability falls below 60%, the system categorises the input as an “Unknown” anomaly and denies access.

8.2 Inference Results

A custom Python inference script was developed to load raw image files from a local directory, apply the necessary ImageNet normalisation transforms, and pass the tensors through the model. The raw logits were then converted to probability percentages.

As visualised in Figure 8, the system successfully verified the novel images of known subjects with extremely high confidence ($> 90\%$). Furthermore, when presented with anomalous data, the confidence scores dropped drastically below the 60% threshold, successfully triggering the “Access Denied” state.

ACCESS DENIED
Unknown / Low Conf
(19.3%)



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Figure 8: Real-world inference on unseen data. The system grants access to recognised users (green) with high Softmax confidence, while thresholding rejects anomalies or unfamiliar scans (red).

This dedicated testing phase confirms that the EfficientNet-B0 pipeline is not merely overfitting to the dataset split, but has genuinely learned the generalisable biometric signatures required for real-world identity authentication.

9 Discussion

9.1 Simple CNN Performance

The simple CNN achieved only 15.56% validation accuracy. This is attributable primarily to the **extreme data scarcity**: with only 4 training images per class (90 classes $\times$ 4 images = 360 total), a CNN trained from scratch has insufficient examples to learn discriminative features. The training curves confirm overfitting — training loss decreases while validation loss increases, indicating that the model memorises rather than generalises.

9.2 MobileNetV2 Performance

The MobileNetV2 transfer learning model achieved 83.33% validation accuracy, a dramatic improvement of 67.78 percentage points. This demonstrates the power of transfer learning in data-limited settings. The pre-trained ImageNet weights provided rich low-level feature representations (edges, textures, shapes) that were directly applicable to the iris recognition task. The model's training curve shows both training and validation accuracy rising steadily together, indicating healthy generalisation without severe overfitting.

9.3 Limitations

- **Dataset size** — only 450 images from 45 subjects is extremely small for a 90-class problem. Larger datasets (e.g., CASIA-IrisV4 with thousands of images) would likely yield significantly higher accuracy.
- **Grayscale images** — iris images are inherently grayscale, while MobileNetV2 expects 3-channel RGB input. Keras handles this by loading images in RGB mode, which may introduce redundant channels.
- **No fine-tuning** — the MobileNetV2 base was kept fully frozen. Unfreezing the top layers (fine-tuning) could further improve accuracy.
- **Class imbalance** — with only 1 validation image per class on average, the validation set is very small and results may have high variance.

9.4 The Power of Unconstrained Fine-Tuning on Large Data

The limitations observed with the MMU dataset were entirely resolved by transitioning to the CASIA combined dataset and utilising two-phase fine-tuning. By forcing the models (ResNet-50 and EfficientNet-B0) to learn across variations in illumination and pupil dilation, the networks learned true, invariant biometric features. The micro-learning rate (1×10^{-5}) used in Phase 2 was crucial; it prevented the destruction of the pre-trained ImageNet weights while allowing the filters to gently adapt to iris textures, culminating in 99% accuracy.

10 Conclusion

This project successfully demonstrated the evolution and application of deep learning for iris-based biometric identification, progressing from a constrained baseline investigation to a highly robust, large-scale system.

The study yielded several critical findings across two distinct phases:

- **Phase 1 (Data-Scarce Environment):** On the small-scale MMU dataset (90 classes), a Simple CNN trained from scratch severely underperformed (15.56% accuracy). However, applying transfer learning via a frozen MobileNetV2 model provided a massive performance boost to 83.33%. This proved that when biometric data is strictly limited, leveraging pre-trained representations is absolutely essential.
- **Phase 2 (Production-Scale Environment):** To achieve state-of-the-art accuracy, the pipeline was scaled up using the PyTorch framework on a dynamically combined CASIA-IrisV4 dataset (18,851 images across 820 unique identities). By implementing a two-phase fine-tuning strategy—where the entire network is unfrozen and trained with a micro-learning rate (1×10^{-5})—the models successfully adapted their pre-trained ImageNet filters to the microscopic textures of the human iris. ResNet-50 achieved an outstanding validation accuracy of **98.99%**, while EfficientNet-B0 peaked at an exceptional **99.12%**.

Ultimately, these results confirm that while neural networks struggle with limited data, the combination of advanced architectural scaling (EfficientNet), robust datasets encompassing real-world environmental variations (illumination and pupil dilation), and unconstrained fine-tuning yields a highly reliable, scalable, and production-ready biometric authentication system. The results confirm a widely observed principle in deep learning: when labelled data is scarce, transfer learning from large pre-trained models is substantially more effective than training from scratch. This has important practical implications for biometric systems, where collecting large labelled datasets is often expensive and privacy-sensitive.

11 Future Work

Several directions could extend and improve this work:

1. **Iris Segmentation** — explicitly isolate the iris region using models like U-Net to crop out eyelashes and eyelids, ensuring the network does not rely on periocular skin features.
2. **Metric Learning** — transition from Softmax classification to Siamese Networks using Triplet Loss. This would allow the system to verify entirely new subjects that were never seen during training (Zero-Shot Learning).
3. **Edge Deployment** — convert the trained PyTorch models to ONNX or TorchScript formats for deployment on embedded biometric hardware or mobile devices.

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